

Evaluating Multimodal Alignment in ChatGPT and Gemini: An Empirical Study of Inter- and Intra-Model Consistency in Image Description and Generation

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Abstract—This small, controlled study examines how well three perceptual metrics—LPIPS, CLIP cosine similarity, and MS-SSIM—align with human preferences over image variants generated by ChatGPT (DALL-E) and Gemini for 15 reference images. Ten raters ranked four variants per reference image. Inter-rater agreement (Kendall’s W) is computed, automatic scores are summarized, and the correlation between human mean ranks and each automatic metric is measured. Humans show moderate agreement ($\bar{W} = 0.40$). Across variants, automatic metrics favor different candidates, and only CLIP shows a modest positive correlation with human ranks on this dataset (mean Spearman 0.36). LPIPS and MS-SSIM correlate weakly. These findings suggest that single off-the-shelf metrics are insufficient proxies for human judgments on this task and that model selection should incorporate targeted human feedback.

Index Terms—human evaluation, perceptual metrics, LPIPS, CLIP, MS-SSIM, alignment

I. INTRODUCTION

Text-to-image systems are typically benchmarked with automatic similarity scores, yet deployment decisions rely on human perception. This study investigates how three widely used metrics align with human preferences over 15 sets of four image variants each. The main contribution is to show, in a small but controlled setting, how three popular metrics diverge from human rankings in a realistic 4-candidate selection scenario, and to quantify intra- vs inter-model consistency for two multimodal LLMs.

Two research questions structure the study:

- RQ1 How consistent, with respect to human preferences, are multimodal AI models in aligning descriptions and generations, both within the same model and across different models?
- RQ2 How well do automatic multimodal evaluation metrics align with human judgments?

Few studies contrast multiple perceptual metrics against ground-truth human orderings for images. By keeping the study small and controlled (15 reference images, fixed four-way choice), the behavior of off-the-shelf metrics is isolated without confounds from model sampling stochasticity. Visual inspection is emphasized alongside quantitative summaries to highlight qualitative failure modes.

II. RELATED WORK

Perceptual similarity has a long history in image quality assessment, from SSIM and MS-SSIM [3], [4] to learned metrics such as LPIPS [1]. CLIP-based similarity is now a common proxy for text–image alignment and aesthetics [2], [5], and recent work trains reward or aesthetic models for human preference modeling. This study complements such work by directly contrasting human rank lists with three popular metrics on the same set of variants, using full-resolution images and strict four-way rankings rather than aggregated thumbnails to reduce ties. Broader surveys on multimodal LLMs and CLIP-based IQA, as well as LLM-based evaluators, outline architectures and automatic preference signals [7]–[10]. Representation alignment (e.g., CLIP embeddings [2]) and cycle-consistency ideas motivate comparing description–generation consistency. In contrast to metric- or model-focused studies, we emphasize comparative analysis of two AI models and their intra-model consistency using both human rankings and automatic metrics on the same bespoke dataset. Dataset-level measures such as FID [6] or aesthetic predictors do not directly address the relative ranking of small candidate sets, so our analysis focuses on the per-image, per-variant view that matches typical human-in-the-loop selection.

III. DATA AND EXPERIMENTAL SETUP

A. Image Variants

Each of the 15 images has a reference version and four model-generated variants (a)–(d). Variants differ in composition and style but share the same prompt intent. Across images, variation is observed in color palette, object placement, and adherence to textual details, providing a useful stress test for perceptual metrics. Table I shows the mapping from letters to generator and caption source. Reference images were collected specifically for this study to minimize the chance that either model encountered them during pretraining (although it is not possible to know whether similar images were in the pretraining data, especially for generic scenes like landscapes and everyday objects). Image selection was manually curated. The small bespoke set spans natural elements (landscapes,

trees, sky, northern lights), embedded text (for instance, on a traffic sign), everyday objects (flags, food, electronics, games), people, and drawings ensuring diversity in content types that stress both compositional and semantic alignment.

TABLE I
MAPPING FROM LETTERS (A)–(D) TO GENERATOR AND CAPTION SOURCE.

	Caption: Gemini	Caption: ChatGPT
Generator: Gemini	(a)	(b)
Generator: ChatGPT/DALL-E	(c)	(d)

B. Human Evaluation

Ten raters (balanced male/female) ranked the four variants for every image. Raters were recruited among students and were blind to model identities. The survey was conducted online. Inter-rater agreement is summarized with Kendall’s W per image and averaged. Raters saw all four variants side-by-side on a laptop screen and selected a strict ordering after brief instructions about prioritizing semantic fidelity and overall appeal. No time limit was imposed. Position of variants was fixed. Participants gave informed consent; no personal data was collected.

C. Automatic Metrics

LPIPS (AlexNet backbone), MS-SSIM, and CLIP cosine similarity (ViT-B/32) are computed. Images are resized to match references for perceptual metrics. Per-variant means and per-image “winners” (best score per metric) are reported.

D. Evaluation Protocol

For each image, raters view the four candidates simultaneously and rank them 1–4. Top-1 rates (fraction of times a variant is ranked first) are aggregated and Kendall’s W is computed to quantify agreement. For automatic metrics, image sizes are aligned to the reference, scores are computed, and variants are ranked per metric and image. Correlations are computed between human mean ranks and metric-induced ranks (scores converted to ranks per image), not between raw scores. Bootstrap confidence intervals and Wilcoxon signed-rank tests in Section IV-D operate on the 15 per-image Spearman coefficients per metric, treating images as independent instances of the task.

IV. RESULTS

A. Human Preferences

Top-1 rates are: (a) (44%), (b) (22%), (c) (13.3%), (d) (20.7%). Inter-rater agreement across all images averages $\bar{W} = 0.40$ (Table III), commonly interpreted as moderate agreement; this loosely upper-bounds how tightly any automatic metric can correlate with the human rankings. Image-level W ranges from low agreement to near-consensus, suggesting image-dependent ambiguity. Fig. 1 summarizes placement percentages across all raters. Given the mapping, raters most often preferred Gemini-generated images prompted by Gemini captions (a), followed by Gemini with ChatGPT

captions (b), and least often ChatGPT-generated images conditioned on Gemini captions (c). The ChatGPT stack with ChatGPT captions (d) lands between (b) and (c), suggesting both the generator and caption source influence perceived quality. The preference table (Table II) shows (a) outranking (b) in 63% of all comparisons, while (c) vs. (d) is nearly balanced (47% vs. 53%), indicating stronger intra-model consistency for Gemini than for ChatGPT. The mean rank-difference heatmap (Fig. 2) echoes this: the a–b row is consistently negative (mean -0.53 ranks), whereas c–d hovers near zero (mean 0.05), with several images flipping order.

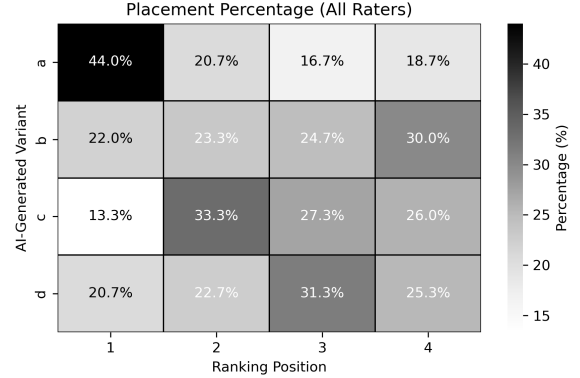


Fig. 1. Human placement percentages across all raters. Variant (a) is most often ranked first.

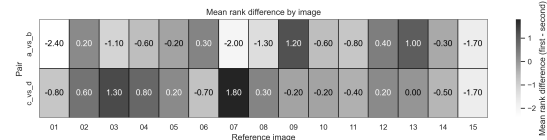


Fig. 2. Mean rank difference (first-second) per image for pairs (a,b) and (c,d). Negative values mean the first variant is ranked higher.

TABLE II
SHARE OF COMPARISONS FAVORING EACH ORDERING WITHIN THE TWO PAIRS (PERCENT WITHIN PAIR).

Pair	$a > b$	$b > a$	$c > d$	$d > c$
Percentage	62.7	37.3	47.3	52.7

TABLE III
PER-IMAGE INTER-RATER AGREEMENT (KENDALL’S W); MEAN $\bar{W} = 0.40$.

	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15
	0.712	0.072	0.388	0.300	0.016	0.396	0.924	0.628	0.348	0.552	0.080	0.028	0.492	0.484	0.580

B. Metric Outcomes

No single metric matches the human-favored variant. Visual inspection confirms that LPIPS and MS-SSIM can reward higher-frequency detail even when humans prefer semantic fidelity, while CLIP occasionally overweights textual align-

ment. Mean metric scores per variant (lower LPIPS is better; higher MS-SSIM/CLIP is better) are reported in Table IV, while Fig. 3 shows how often each variant wins per metric. When counting per-image wins (global percentage over all rows), LPIPS favors (a) (11.7%) while CLIP favors (a) (10.0%) and (d) (8.3%); MS-SSIM splits wins across (b)–(d) (8.3% each) with minimal wins for (a) (1.7%). LPIPS and CLIP lean toward the Gemini-generated images prompted by Gemini captions variant (a), whereas MS-SSIM spreads wins across Gemini-generated images prompted by ChatGPT captions (b) and ChatGPT-generated images prompted by ChatGPT captions (d), indicating that neither Gemini nor ChatGPT pipelines dominate across perceptual criteria. The spread reinforces that off-the-shelf metrics disagree on whether Gemini-generated or ChatGPT-generated images better match references, even when fed the same caption source. Fig. 4 contrasts human top-1 rates with global metric win percentages to highlight where automatic selection diverges from human preference. Across metrics, the mean scores in Table IV lie in a narrow band, implying that automatic measures view the variants as roughly similar, whereas human preferences are more polarized toward variant (a), underscoring a gap between numerical similarity and subjective choice.

C. Rank Distributions

Beyond single winners, the typical rank positions of each variant are examined when ordered by a metric. LPIPS places variant (a) first most often but also pushes it to lower ranks in some cases. MS-SSIM distributes ranks more evenly, while CLIP concentrates wins on (a) and (d). These distributions echo the weak correlations with human preferences: no metric consistently mirrors the human ordering.

TABLE IV
MEAN AUTOMATIC SCORES PER VARIANT (LOWER LPIPS IS BETTER;
HIGHER MS-SSIM/CLIP IS BETTER).

Variant	LPIPS	MS-SSIM	CLIP cosine
(a)	0.743	0.399	0.788
(b)	0.764	0.406	0.773
(c)	0.755	0.458	0.786
(d)	0.764	0.443	0.790

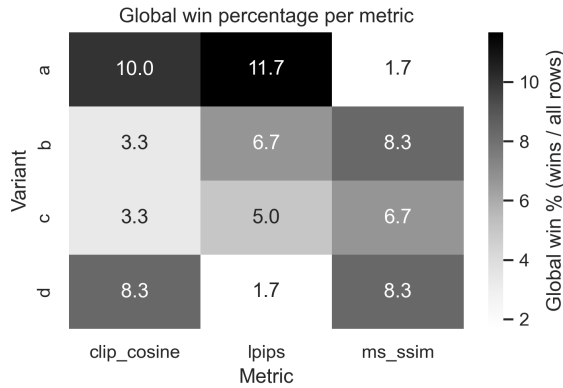


Fig. 3. Global percentage of per-image wins by variant and metric.

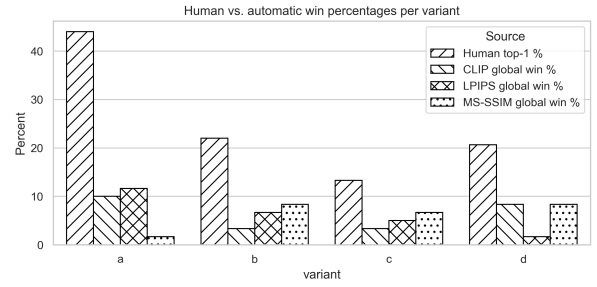


Fig. 4. Human top-1 rates compared to automatic global win percentages per variant.

D. Correlation Analysis

Table V reports the mean correlation between human mean ranks and each metric over 15 images. CLIP has the highest mean Spearman (0.36); all bootstrap CIs include zero, but CLIP's interval is shifted upward (0.36 [−0.02, 0.67]) relative to LPIPS (0.11 [−0.25, 0.46]) and MS-SSIM (0.06 [−0.29, 0.40]; Fig. 5). This is consistent with a modest positive relationship for CLIP but a sample too small to be certain, while LPIPS and MS-SSIM remain fully compatible with no real correlation. A one-sided (directional) Wilcoxon signed-rank test on per-image Spearman values finds CLIP greater than LPIPS ($p=0.13$) and greater than MS-SSIM ($p=0.07$), indicating only weak evidence that CLIP outperforms the other metrics in this small- N setting. Some images exhibit negative or near-zero correlation, indicating cases where metrics are anti-aligned with human preference.

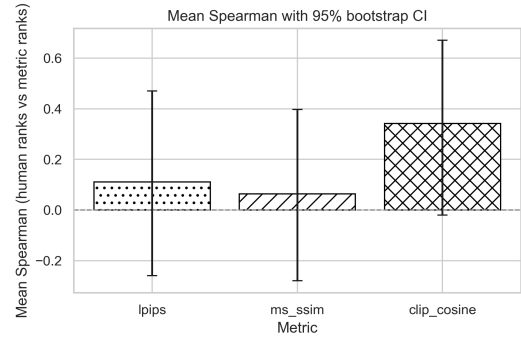


Fig. 5. Mean Spearman (human mean ranks vs. metric ranks) with 95% bootstrap CIs.

TABLE V
MEAN CORRELATIONS (15 IMAGES) BETWEEN HUMAN MEAN RANKS AND
AUTOMATIC METRICS.

Metric	Kendall	Spearman	Pearson
CLIP cosine	0.32	0.36	0.35
LPIPS	0.12	0.11	0.13
MS-SSIM	0.01	0.06	0.10

V. DISCUSSION

Humans prefer variant (a) nearly twice as often as (b) and three times as often as (c), yet automatic metrics do not

unanimously select (a). CLIP tracks human rankings better than LPIPS or MS-SSIM on average, although this advantage is only weakly supported statistically in our small sample (Wilcoxon $p > 0.05$), and substantial disagreement remains. Moderate inter-rater agreement ($\bar{W} = 0.40$) further limits how strongly any metric can align, suggesting that image difficulty and subjective taste both contribute noise. Since both $\bar{W}$ and the metric–human correlations are computed per image and then averaged, each image acts as an independent instance of the task, and disagreement between raters sets an upper bound on how coherent any metric–human relationship can be on this dataset. The inconsistency across metrics underscores the need for either metric ensembling, human-in-the-loop ranking, or task-specific fine-tuning. Expanding rater diversity and scale is future work. Practical guidance: if constrained to a single metric, CLIP is the least misaligned here; otherwise, collect a small human validation set and calibrate a weighted combination.

A. Limitations

The study covers only 15 images and four variants per image, limiting statistical power and diversity. Rater demographics are narrow (ten participants), so subgroup analysis is indicative only. All metrics are used off-the-shelf; better alignment may be achievable with fine-tuned or task-specific preference models. Images are evaluated at a single resolution and device configuration; variants were shown in fixed positions, which may introduce mild positional bias, and different scales or hardware could modestly shift scores.

B. Practical Implications

For small-batch creative workflows where a designer must pick one of several candidates, these pilot results (15 images, $\bar{W} = 0.40$) suggest that a semantic signal such as CLIP may be a safer default than purely perceptual scores, but this is a hypothesis rather than a validated prescription. A pragmatic design idea is to pair one semantic metric with a quick human pass, or to learn metric weights on a small, task-specific human calibration set; both would need targeted testing. In automated pipelines, exposing multiple candidates to end users may be preferable to auto-selecting a “best” image based solely on any single off-the-shelf metric.

C. Future Directions

Expanding the image set and rater pool would allow finer-grained subgroup analysis and statistical testing. Per-image, per-rater randomization of positions could reduce potential positional bias. Prompt engineering is another avenue: future experiments can explicitly provide models with information about the target audience and task purpose when generating captions or images. For captioning, the audience cue can specify whether the consumer is the same model that will later generate images or a different AI model. For generation, prompts can state whether the output is destined for human ranking or an automatic tool, enabling evaluation of how audience- and purpose-aware instructions affect alignment.

VI. CONCLUSION

This study collected 15 bespoke reference images with four variants each, generated by Gemini and ChatGPT using both self-authored and cross-authored captions, to minimize pretraining leakage and diversify content (people, objects, text, nature). Ten human raters provided a ranking of the four variants for each reference image, yielding moderate inter-rater agreement ($\bar{W} = 0.40$). Automatic evaluation used LPIPS, MS-SSIM, and CLIP to score and rank the same variants, enabling a direct comparison between human preferences and widely used perceptual metrics.

RQ1 Human raters favored Gemini-generated images prompted by Gemini captions (variant (a)), with lower preference for ChatGPT-generated images; intra-model consistency is stronger than inter-model consistency, since each model paired with its own captions ((a) and (d)) is ranked first more often than when using the other model’s captions ((b) and (c)). Table II and Fig. 2 highlight this: (a) beats (b) in 63% of pairwise comparisons and is, on average, half a rank better, while (c) vs. (d) is nearly split with small mean differences, suggesting that Gemini is less sensitive to caption source than ChatGPT.

RQ2 CLIP shows the strongest (yet modest) alignment with human rankings, while LPIPS and MS-SSIM correlate weakly, so no single automatic metric adequately proxies human judgment.

Across metrics, mean scores cluster tightly, suggesting automatic measures perceive variants as similar, whereas human choices are more polarized toward (a). Selecting variants solely by one metric therefore risks diverging from user preferences. All code for metric computation and analysis, together with anonymized images and rankings, is available at <https://github.com/SaverioNapolitano/qrd>. Future work should expand the human study, explore aesthetic and learned preference models, test metric combinations tuned to human-labeled data, and further probe audience- and purpose-aware prompting to improve downstream alignment.

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